

Andrew Yonan (10973951)

Department of Applied Mathematics and Statistics

COLORADO SCHOOL OF MINES

MATH 455: Partial Differential Equations

Assignment #2 - More Fourier Series

Due Thursday, September 17, 2026

1. (12 points) Define the function

$$f(x) = \begin{cases} 1, & \text{if } |x| < \frac{\pi}{2} \\ 0, & \text{if } |x| > \frac{\pi}{2}, \end{cases}$$

and let $F(x)$ be the Fourier series of $f(x)$ on the interval $(-\pi, \pi)$. Sketch by hand $F(x)$ on the interval $(-3\pi, 3\pi)$ and compute a formula for $F(x)$ using the Fourier coefficients.

Solution — Since f is not periodic on $\mathbb{R}$, we restrict its domain to $[-\pi, \pi]$ and fill in left-right-averaged discontinuities:

$$f_{2\pi}(x) = \begin{cases} 1 & \text{if } |x| < \frac{\pi}{2} \\ 0 & \text{if } \frac{\pi}{2} < |x| \leq \pi \\ \frac{1}{2} & \text{if } |x| = \frac{\pi}{2} \end{cases}$$

Let $g(x)$ be the periodic extension of $f_{2\pi}(x)$ to all $\mathbb{R}$. By the Fourier convergence theorem, $F(x) = g(x)$ for all $\mathbb{R}$, so sketching $F(x)$ on $(-3\pi, 3\pi)$ is equivalent to sketching $g(x)$ on $(-3\pi, 3\pi)$:

2. (8 points) We know that if $f(x)$ is odd on the interval $(-\pi, \pi)$, then its Fourier series is composed of only sine functions. What additional symmetry condition on f will make the sine coefficients with even indices become zero (i.e., $b_{2n} = 0$ for all $n \in \mathbb{Z}$)? Provide a specific example of a function with this property and show that it satisfies the condition.

Solution — A

3. (12 points) Let

$$\begin{aligned} f(x) &= x \cos(x), & x &\in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \\ g(x) &= x \cos(x), & x &\in (-1, 1), \end{aligned}$$

and

$$h(x) = \begin{cases} 0, & \text{if } 1 < x < 3 \\ 1, & \text{if } -1 < x < 1 \\ x, & \text{if } -3 < x < -1 \end{cases}$$

be periodic functions (defined via periodic extension) for $x \in \mathbb{R}$. For each function

- (a) Determine whether or not the function is in PC^1 on the interval over which it is defined.

Solution — A

- (b) State the function to which its Fourier series (defined on the interval: $[-\frac{\pi}{2}, \frac{\pi}{2}]$ for $f(x)$, $[-1, 1]$ for $g(x)$, and $[-3, 3]$ for $h(x)$) converges

Note: Do not compute the Fourier series (or coefficients) to do this!

Solution — A

- (c) Sketch by hand three periods of the Fourier series in part (b).

Solution — A

4. (8 points) Let

$$F(x) = \sum_{n=1}^{\infty} \frac{(-1)^n n^{3/2} + 2}{n^8 + 2n^6 + 1} \cos(nx)$$

How many continuous derivatives can we be sure that F possesses? Justify your answer using tools from Calculus II.

Solution — A